Lending Business Data Analysis

This project analyzes a synthetic dataset representing a short-term consumer lending (quick loan) portfolio. The goal is to explore loan performance, borrower characteristics, and financial risk indicators to better understand lending decisions and portfolio structure.

Data Inspection

After importing the dataset, I examined it using .describe() and missing value checks to verify data completeness and understand variable ranges.

The dataset contains the following variables:

- **loan_id** — unique identifier of the financing application (and loan if approved)
- **application_date** — date the financing request was submitted / activated
- **requested_amount** — amount requested and issued if approved
- **term_days** — loan duration in days
- **total_repayment_expected** — total scheduled repayment amount
- **total_repayment_received** — total amount repaid to date
- **last_payment_date** — date of most recent repayment
- **approval_flag** — 1 if approved, 0 otherwise
- **customer_id** — unique customer identifier
- **customer_birth_date** — date of birth
- **verified_monthly_income** — verified net monthly income
- **verified_monthly_expenses** — verified monthly living expenses
- **external_credit_commitments** — ongoing repayment obligations to other lenders
- **overdue_external_balance** — overdue external debt amount

Missing Values

Missing values appear only in **last_payment_date**.

This is expected and reflects business logic — a payment date exists only when a repayment has been made. Loans without repayment (not yet paid or not approved) correctly have no payment date.

Therefore, missing values are structural and do not require treatment.

Data Preparation

A subset of approved loans was created for certain analyses.
No other structural changes were made to the dataset.

Exploratory Data Analysis (EDA)

KPI Analysis

Key aggregated portfolio metrics show:

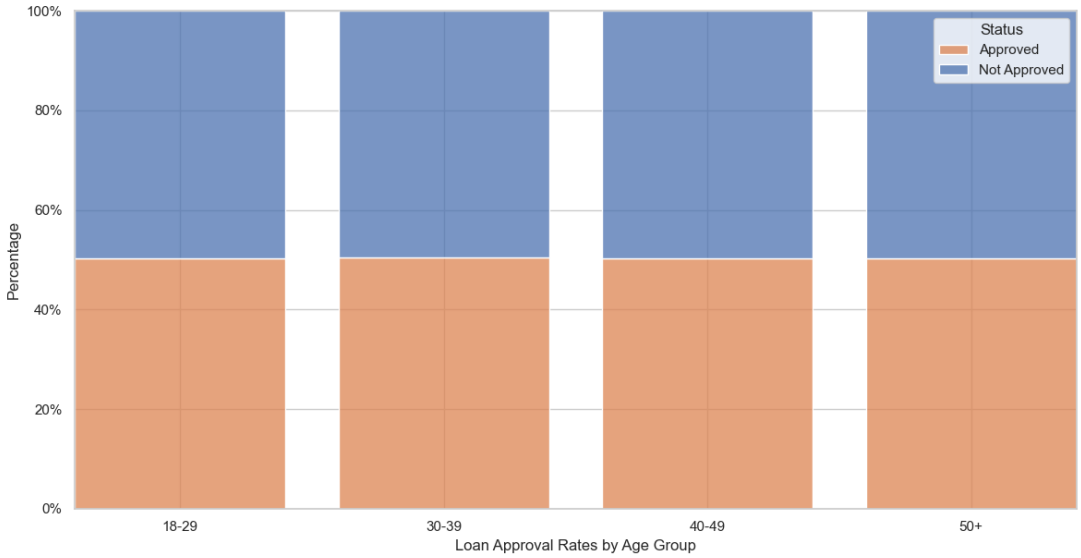
- The average loan term confirms a short-term lending product.
- Requested amounts range up to €5,000, consistent with quick-loan / micro-credit positioning.
- Total outstanding balance reflects unpaid receivables across the portfolio.
- Expected income represents total contractual repayment above principal.
- Realized income is currently negative because many loans are not fully repaid yet or only partially repaid.

This indicates the portfolio is still in repayment phase and some loans may default or remain outstanding.

Client Age

Applicants were grouped into age segments and compared by approval status.

Approval rates are broadly similar across age groups, indicating that age alone does not strongly influence lending decisions. The borrower base is primarily working-age individuals, typical for short-term credit products.



Debt-to-Income Ratio (DTI)

DTI is calculated as overdue external balance ÷ verified monthly income.

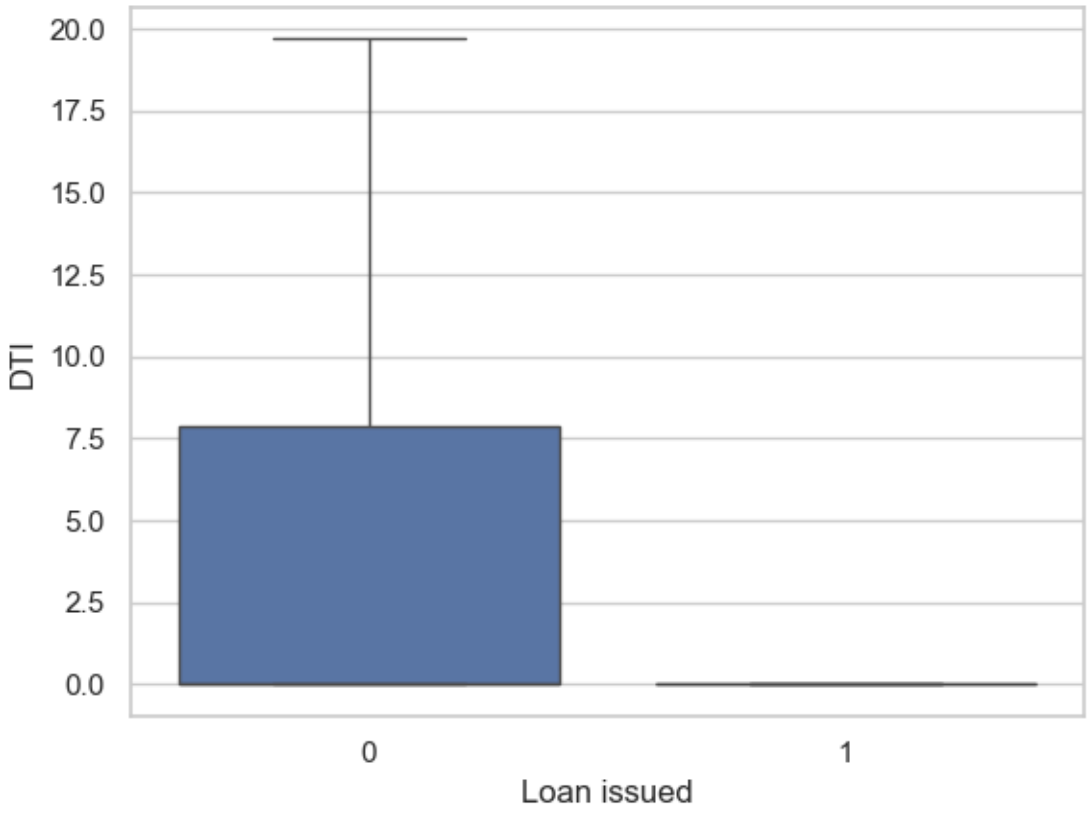
This reflects the share of income represented by **past-due external debt**, not total debt.

Findings:

- Median DTI is 0% for both approved and rejected applicants, meaning most borrowers have no overdue external debt.
- Average DTI is higher among rejected applicants (8.76%) than approved applicants (5.32%).
- 4% of approved clients have DTI above 40%, representing a higher-risk segment.

The difference between mean and median indicates a highly skewed distribution — most borrowers have zero overdue debt, while a smaller group carries large overdue balances.

Boxplot analysis shows rejected applicants generally have broader and higher DTI ranges, suggesting approvals are more common among lower-DTI borrowers.



Expense-to-Income Ratio (EIR)

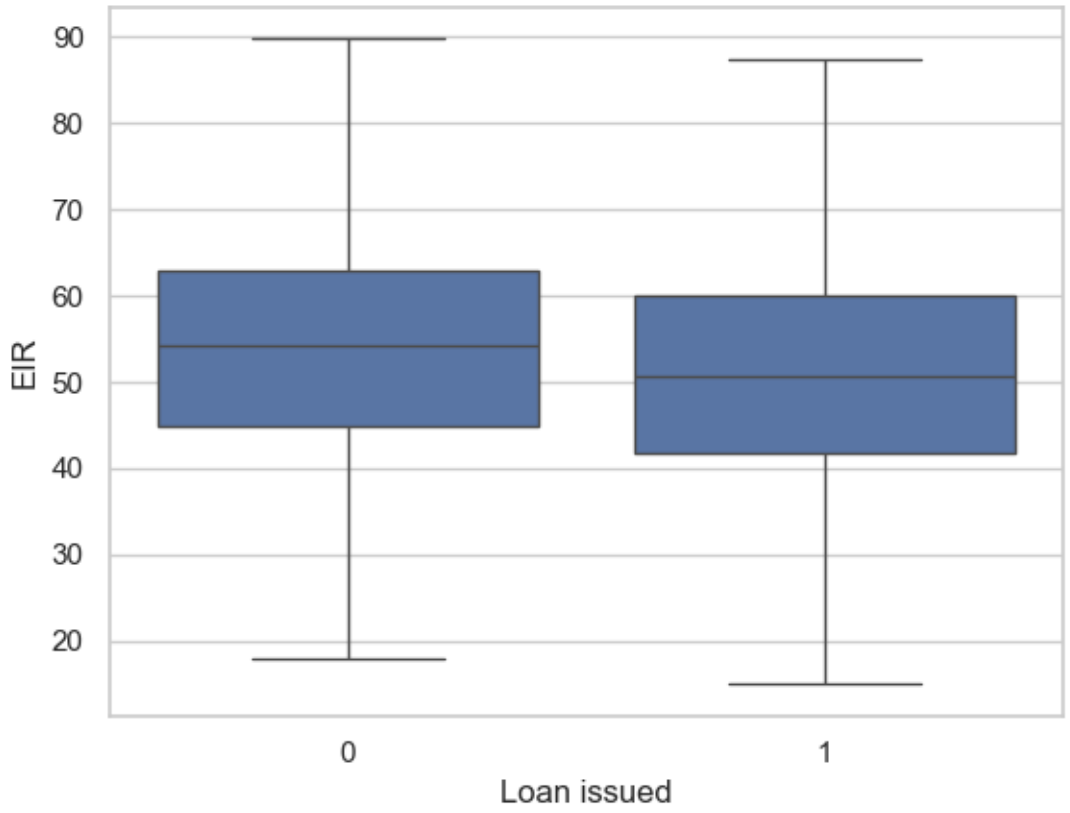
EIR is calculated as verified monthly expenses ÷ verified monthly income.

This measures the share of income required for regular living costs.

Findings:

- Approved applicants show slightly lower average EIR (50.99%) than rejected applicants (53.92%).
- Median EIR is also lower among approved borrowers.
- 52% of approved clients have EIR above 50%, indicating many borrowers already use a large share of income for expenses.

This suggests expense burden affects affordability but is not a strict approval threshold.

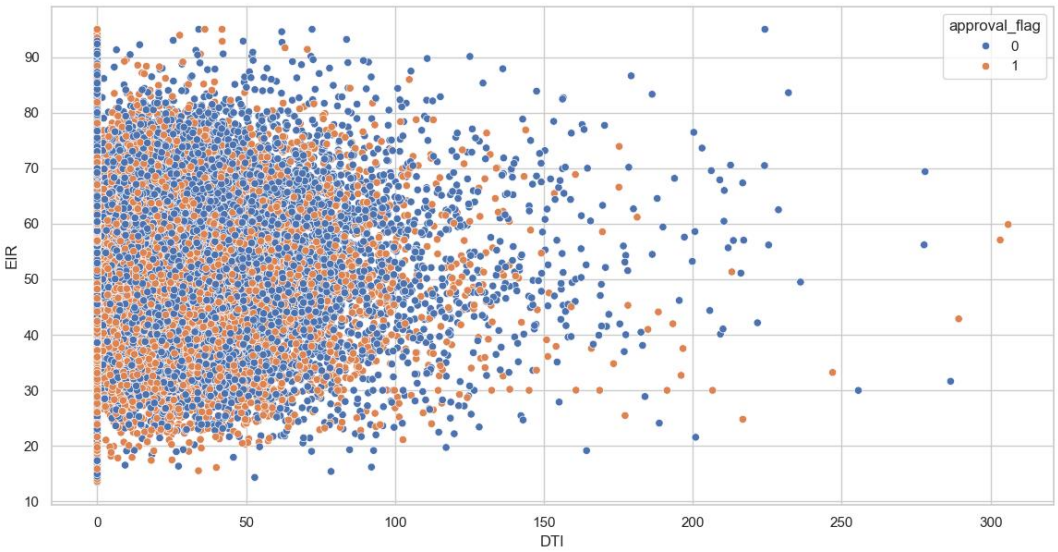


Client Risk Profile: DTI vs EIR

A scatterplot comparing DTI and EIR shows substantial overlap between approved and rejected applicants.

While higher ratios indicate financial strain, approval decisions are not determined solely by these metrics. Lending decisions likely incorporate additional variables such as income level, loan size, or other risk indicators.

Some high-burden applicants are still approved, indicating moderate risk exposure within the portfolio.



Key Findings

- **Short-term lending model** — loan durations are brief and designed for immediate financing needs.
- **Small loan sizes** — typical quick-loan structure.
- **Majority of borrowers have no overdue external debt**, but a smaller high-risk segment exists.
- **Expense burden is substantial for many borrowers**, limiting repayment capacity.
- **Approval decisions are multi-factor**, not driven solely by DTI or EIR.
- **Portfolio still in repayment phase**, resulting in currently negative realized income.

Business Considerations

Potential strategies to improve profitability:

Refine risk assessment

- Pros: better repayment performance
- Cons: fewer approved borrowers

Adjust pricing for risk segments

- Pros: compensates for higher risk
- Cons: may increase default probability

Extend repayment terms with lower periodic burden

- Pros: improved affordability and repayment stability
- Cons: longer capital recovery time

Overall Insight

The portfolio serves primarily short-term borrowers with moderate to high financial burden. While most clients show no overdue external debt, expense levels are relatively high, and approval differences across risk indicators are modest.

This suggests a lending strategy that balances growth with risk exposure, but long-term performance depends on repayment realization across currently outstanding loans.